Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 9 - Week 9 (March 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

What specific limitation of Bitcoin-NG does ByzCoin attempt to solve?

a. The requirement for all miners to share their private keys with the entire network
b. The introduction of faulty or malicious microblocks that are only discovered at the end of a round
c. The inability to elect a leader using proof of work
d. The lack of support for any form of cryptographic signatures on transactions

Answer: (b)

Detailed solution:
In Bitcoin-NG, a leader can inject faulty microblocks that are only invalidated once a new key block is mined, causing significant overhead.

QUESTION 2

The 'Blockchain Performance Triangle' illustrates the trade-offs between which three properties?

a. Throughput, Latency, and Scalability
b. Scalability, Security, and Decentralization
c. Consistency, Availability, and Tolerance
d. Proof of Work, PBFT, and Collective Signing.

Answer: (b)

Detailed solution:
Please refer to week 9 lecture 41.

QUESTION 3

ByzCoin improves upon traditional Practical Byzantine Fault Tolerance (PBFT) by integrating it with Proof of Work (PoW) and Collective Signing (CoSi). Which of the following statements correctly describe the technical improvements in ByzCoin for a network of n nodes?

a. ByzCoin reduces communication complexity from $O(n^2)$ to $O(n)$ by using a tree-based multicast protocol.
b. ByzCoin reduces consensus latency from $O(n^2)$ to $O(\log n)$ by using a tree-based communication structure.
c. ByzCoin reduces verification complexity for third-party observers from $O(n)$ to $O(1)$ by using Schnorr multi-signatures.
d. ByzCoin eliminates the need for a leader node, reducing the communication complexity to $O(1)$.

Answer: (b) and (c)

Detailed solution:
Standard PBFT requires $O(n^2)$ messages because every node communicates with every other node. ByzCoin uses a tree-based multicast (CoSi), which reduces the communication complexity (total messages) to $O(n)$. Because the messages travel down the height of the tree, the latency is reduced to $O(\log n)$. Furthermore, ByzCoin utilizes Schnorr multi-signatures to aggregate n signatures into a single collective signature, allowing for $O(1)$ verification complexity by any entity.

QUESTION 4

To select a random committee without centralized coordination, Algorand uses **Cryptographic Sortition** via a Verifiable Random Function (**VRF**). When a user runs the VRF, it outputs a hash, a proof, and an integer j. If the algorithm returns j=0 for a user, what does this mathematically signify?

a. The user is selected as the sole block proposer for the current round.
b. The user is a part of the committee, but with a voting weight of 0.
c. The user is not a part of the consensus committee for that round.
d. The VRF failed, and the user must re-run the algorithm using the seed from round $r-1$.

Answer: (c)

Detailed solution:

The VRF function executed by every node during cryptographic sortition returns a hash, a proof, and a value j. If j is non-zero, the user is a part of the committee. If j = 0, the user is not part of the committee.

QUESTION 5

Under Algorand's synchrony assumptions, what characterizes **weak synchrony**?

a. There is no communication between nodes
b. Messages are guaranteed to be received within a known time bound by almost all honest users.
c. The network can be controlled by an adversary for a bounded period of time.
d. The protocol only works if the majority of the money is held by one central bank.

Answer: (c)

Detailed solution:
Weak synchrony assumes that while an adversary might partition the network or delay messages, they can only do so for a limited, finite duration before strong synchrony resumes.

QUESTION 6

Which of the following is NOT true about Single Sign-on?

a. A single user can access multiple services.
b. A single identity provider typically maintains the identity of an individual
c. During authentication, the identity is not exposed to all the services
d. Identity holders use the Identity Provider to authenticate other identity holders

Answer: (d)

Detailed solution:

Please refer to Week 9 lecture 43.

QUESTION 7

Decentralized Identifiers (DIDs) are a W3C recommendation to implement self-sovereign identity. Which of the following accurately describes the technical components and structure of DIDs?

a. A DID URI typically follows the syntax: `did:<method-name>:<method-specific-identifier>`.
b. The DID document must contain only JSON-encoded data to ensure compatibility with all blockchain-verified registries.
c. The DID document includes cryptographic public keys and authentication methods so the verifier can verify that the subject controls the DID.
d. A DID document specifies a "service endpoint," which tells the verifier how to communicate with the DID subject for the next level of service.

Answer: (a), (c), and (d)

Detailed solution:
A DID URI consists of the scheme (did), method, and method-specific identifier. A DID Document contains cryptographic public keys for authentication and service endpoints to define how to communicate with the subject. Statement B is false because DID Documents are flexible and can use representations other than JSON (e.g., XML), defined in representation-specific entries.

QUESTION 8

In the Verifiable Credentials (VC) data model, which entity is responsible for asserting claims about a subject and cryptographically signing them?

a. The Holder
b. The Verifier
c. The Issuer
d. The Blockchain miner.

Answer: (c)

Detailed solution:
The issuer, like a university, is the authority that creates the VC, signs it with a private key, and gives it to the holder.

QUESTION 9

If Algorand's BA* protocol fails to reach a final consensus in a single round, how does it resolve the situation mathematically and architecturally?

a. It runs the BA* protocol repeatedly, then a timeout occurs, and the protocol moves to the next round with a newly selected random committee.
b. Mathematically, exactly 10 executions of BA* are required to force a final consensus under weak synchrony.
c. Algorand mathematically demonstrates that running BA* 2 to 3 times is typically sufficient to reach final consensus, even if up to 49% of participants are malicious.
d. It permanently halts the blockchain to prevent a fork, requiring off-chain governance to reset the verifiable random functions.

Answer: (a) and (c)

Detailed solution:
To escape tentative consensus, Algorand triggers a timeout and moves to the next round of BA* with an entirely new, randomly selected committee chosen via cryptographic sortition. The mathematical proof behind Algorand shows that running BA* 2 or 3 times is generally sufficient to reach a final consensus, even with up to 49% malicious actors

Please also refer to week 9 lecture 42.

QUESTION 10

In the structure of a Decentralized Identifier (DID) URI such as `did:example:123`, what does 'example' represent?

a. The DID method
b. The DID subject identifier
c. The DID document
d. The DID scheme

Answer: (a)

Detailed solution:
The second component of the URI specifies the method that supports the DID scheme and defines how the identifier is resolved.